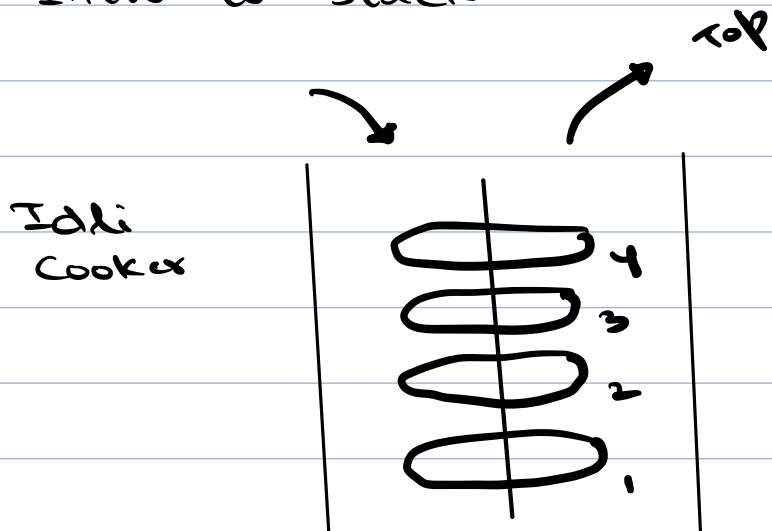


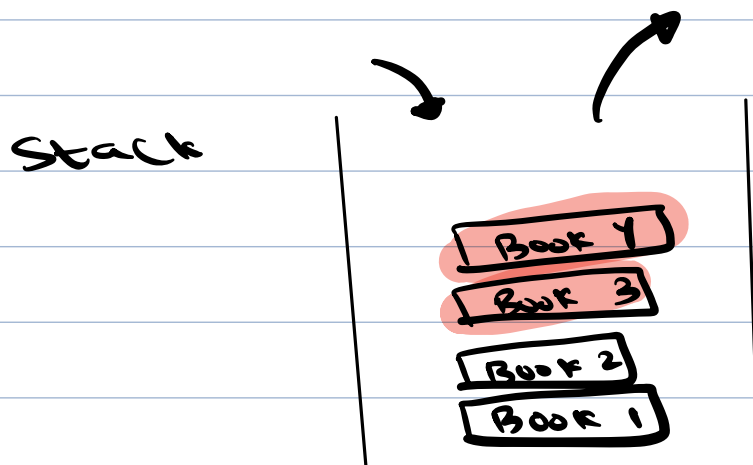
Intro to Stack



1. can insert and remove from top
2. only have access to the top plate

LIFO

(Last In First Out)



1. can insert and remove from top
2. only have access to the top element

Top: Book 4

Remove: Book 4

Remove: Book 3

Top: Book 2

Fn call stack

Call Stack

Ex 1

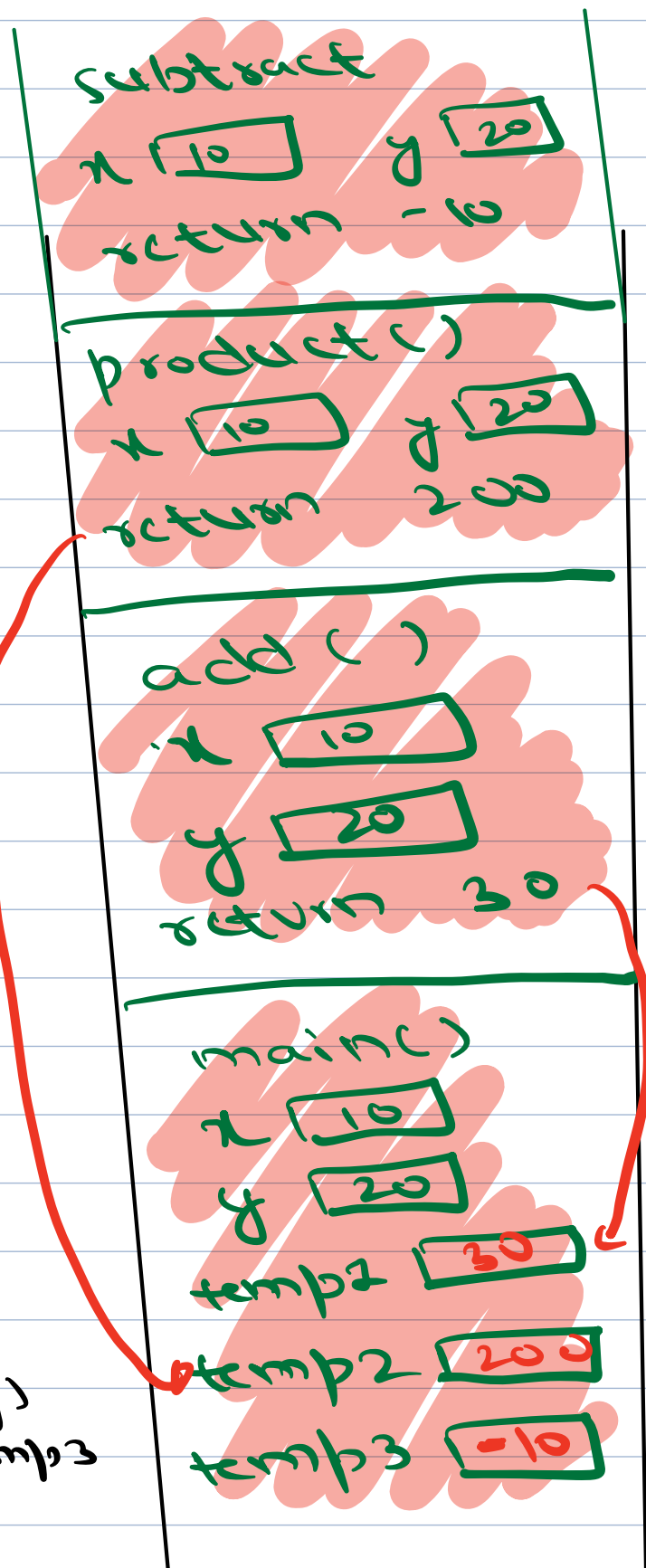
```
int add (int x, int y) <
| return x+y
>
```

```
int product (int x, int y) <
| return x*y
>
```

```
int subtract (int x, int y) <
| return x-y
>
```

```
public static void main() <
| int x = 10
| int y = 20
| int temp1 = add (x, y)
| int temp2 = product (x, y)
| int temp3 = subtract (x, y)
| print (temp1 + temp2 + temp3)
>
```

O/P : 220



When a function is called $\rightarrow$ put on stack
Top most function executes

Ex 2

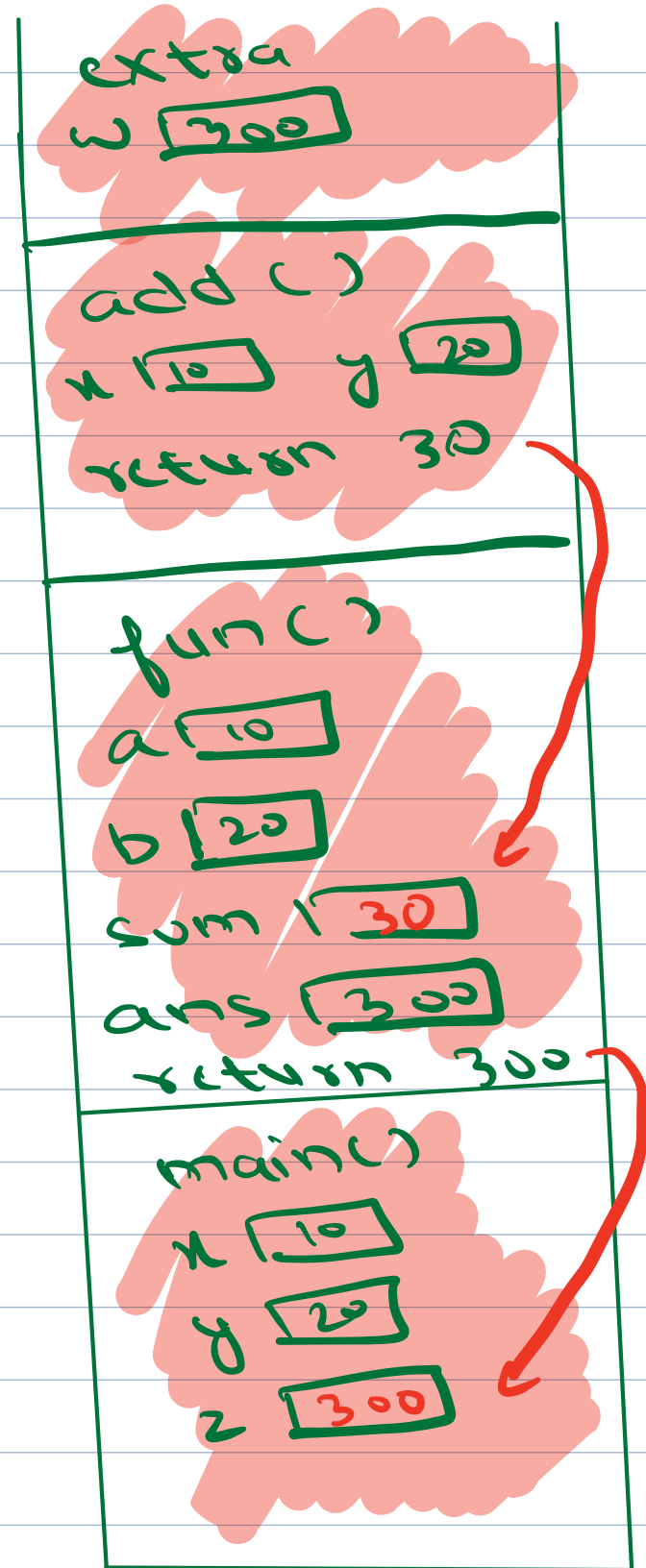
```
int add (int x, int y) <
| return x+y
>
```

```
static int fun (int a, int b) <
| int sum = add (a, b)
| int ans = sum * 10
| return ans
>
```

```
static void extra (int w) <
| print ("Hello")
| print (w)
>
```

```
public static void main () <
| int x = 10
| int y = 20
| int z = fun (x, y)
| print (z)
| extra (z)
>
```

O/P : 300
Hello
300



Fn call stack

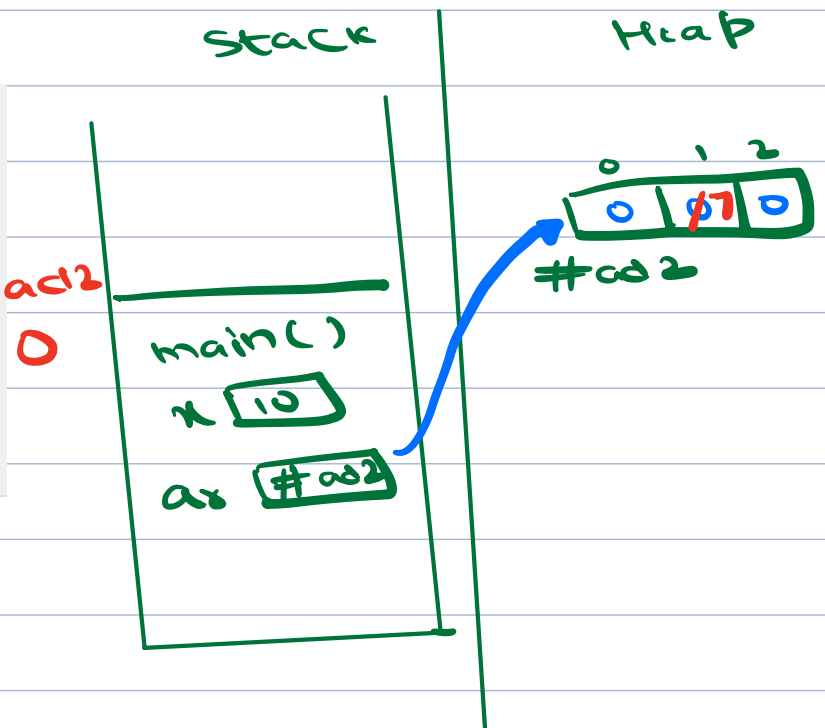
Types of Memory -

1. Stack - All primitive data type (int, float, char, double, boolean, long)
Reference / address of the container
2. Heap - Containers of that reference
(Array / ArrayList / objects)

Reference variable

It is a variable which stores address to data (another variable)

```
public static void main() {  
    int x = 10;  
    int[] ar = new int[3];  
    System.out.println(ar); #ad2  
    System.out.println(ar[2]); 0  
    ar[1] = 7;  
}
```



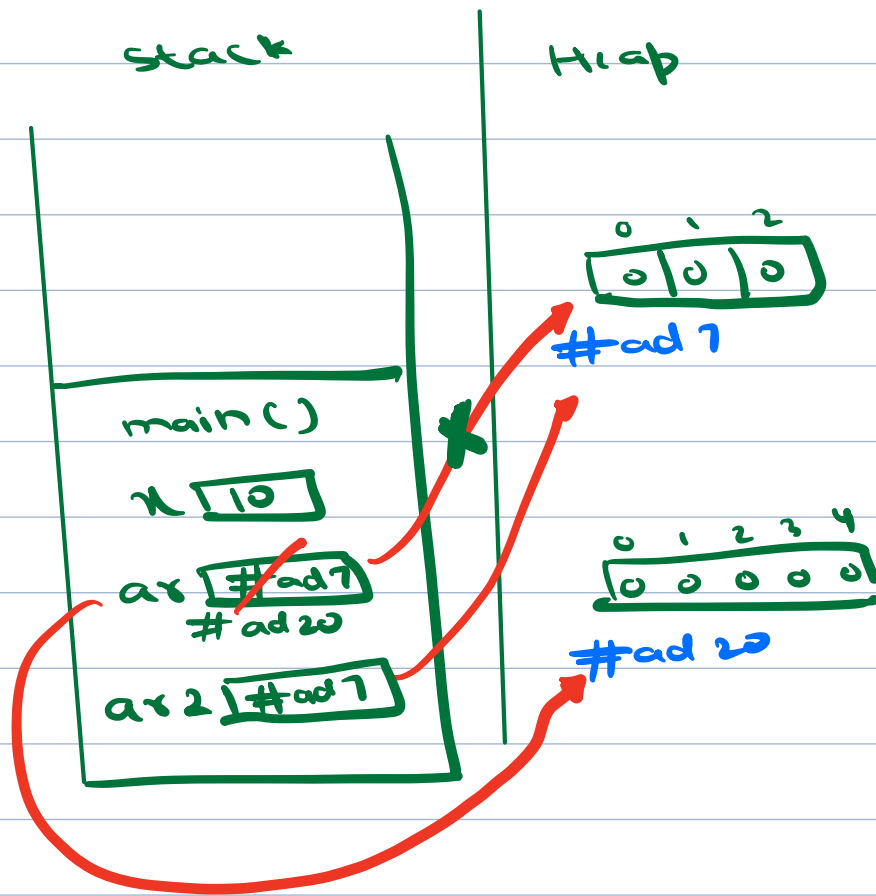
int[] a

↓
reference var which will store
starting address of int[]

```
public static void main() {
    int x = 10;
    int[] ar = new int[3];
    int[] ar2 = ar;
    System.out.println(ar);
    System.out.println(ar2);
}
```

ar = new int[5]
println(ar)

O/P: ad7 (ar)
ad7 (ar2)
ad20 (ar)



PASS BY REFERENCE

`int[] a` → 'a' is a reference variable storing address of integer array

```

static void fun(int[] a){
    System.out.println(a);
    a[1] = 5;
}

public static void main() {
    int[] ar = new int[3];
    System.out.println(ar);
    ar[0] = 90;
    ar[1] = 50;
    fun(ar);
    System.out.println(ar[1])
}

```

ad7

ad7

5

Stack

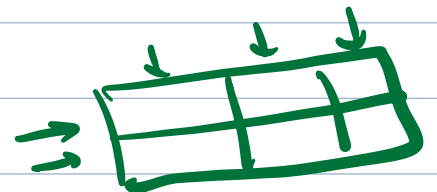
Heap



int mat[4][2]

Every row → 2D array

Matrix | 2D array



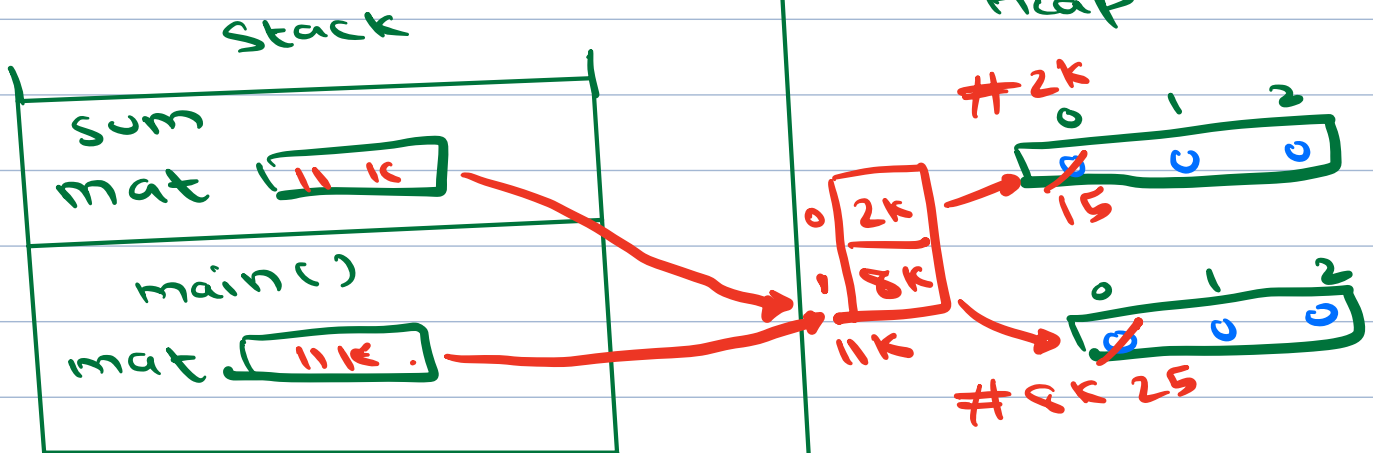
```
static void sum(int[][] mat){
    System.out.println(mat); ██████ 11 K
    System.out.println(mat[0][0] + mat[1][0]); 40
    15 + 25
}

public static void main() {
    int[][] mat = new int[2][3];
    mat[0][0] = 15;
    mat[1][0] = 25;
    sum(mat);
}
```

mat [2][3]
rows cols

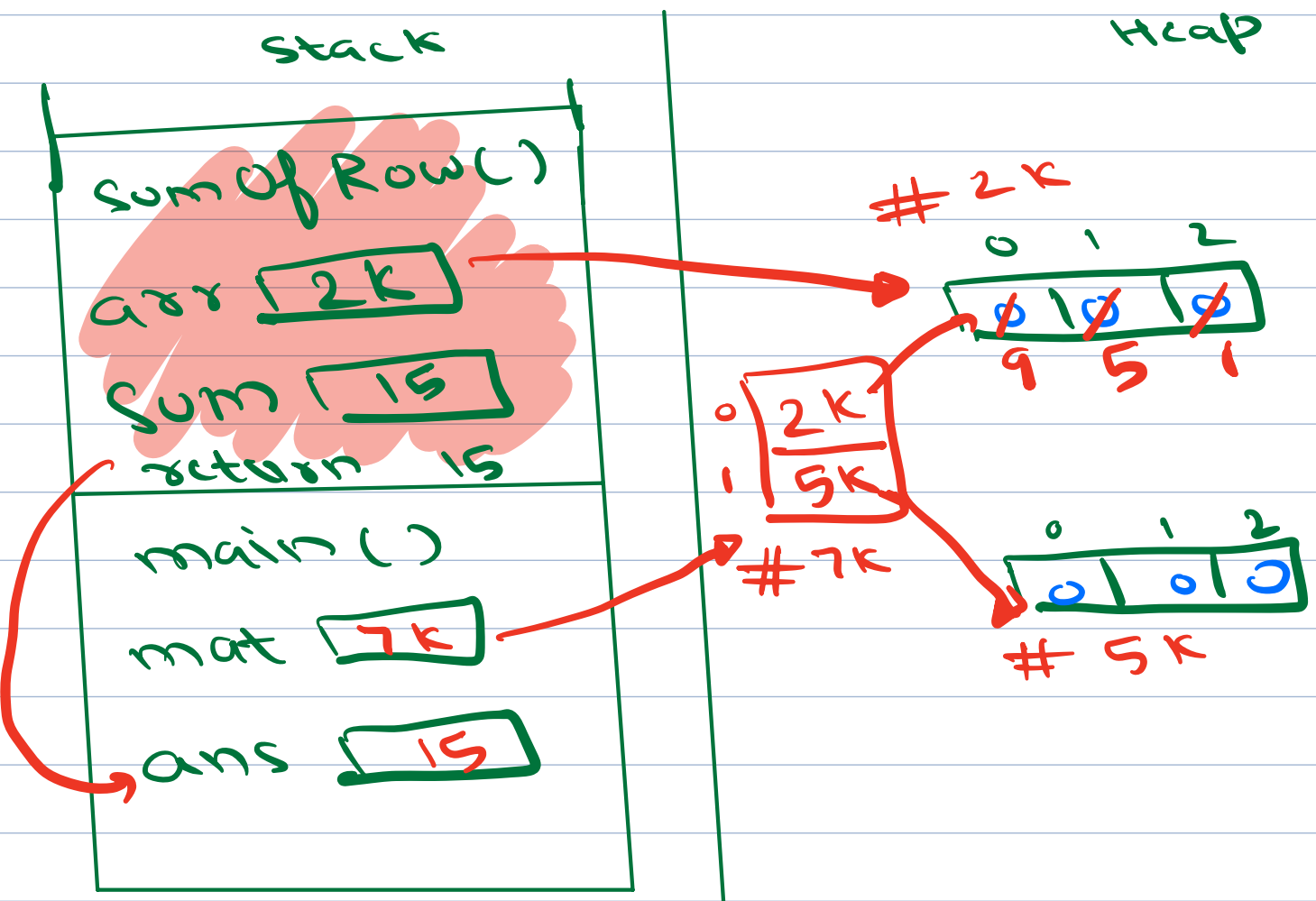
mat → 11 K
mat[0] → 2 K
mat[1] → 8 K

Array of references



```
static int sumOfRow(int[] arr){
    System.out.println(arr); ██████
    int sum = 0;
    for (int i = 0; i < arr.length; i++){
        sum = sum + arr[i];
    }
    return sum;
}
```

```
public static void main() {
    int[][] mat = new int[2][3];
    mat[0][0] = 9;
    mat[0][1] = 5;
    mat[0][2] = 1;
    int ans = sumOfRow(mat[0]); ██████
    System.out.println(ans); ██████ 15
}
```



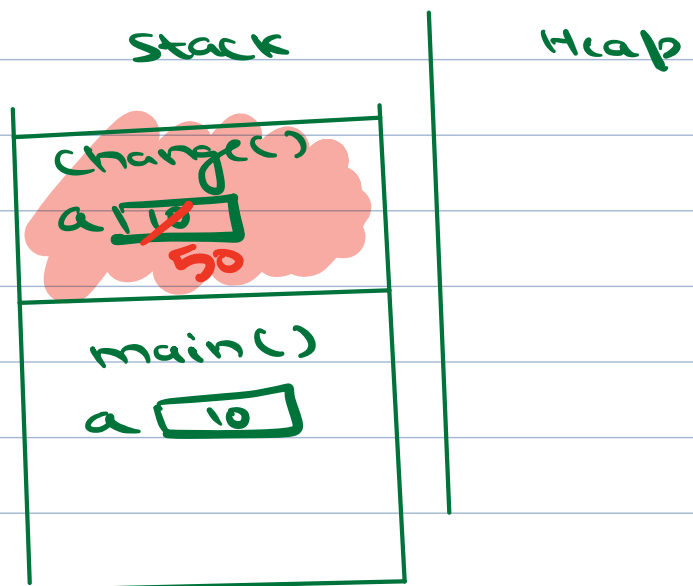
Quiz 2

```

static void change(int a) {
    a = 50;
}
  
```

```

public static void main(String args[]) {
    int a = 10;
    change(a);
    System.out.println(a); 10
}
  
```

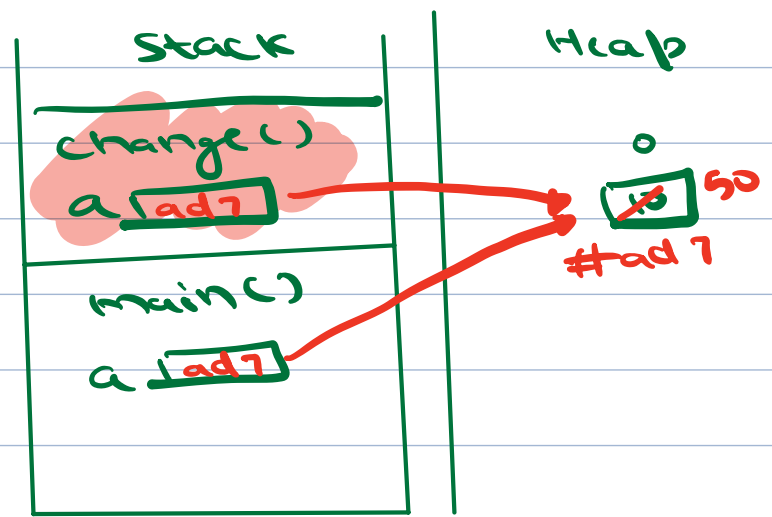


① PASS BY VALUE

② stack changes are temporary

Quiz 2

```
static void change(int[] a) {  
    a[0] = 50;  
}  
  
public static void main(String args[]) {  
    int[] a = {10};  
    change(a);  
    System.out.println(a[0]); 50  
}
```



①

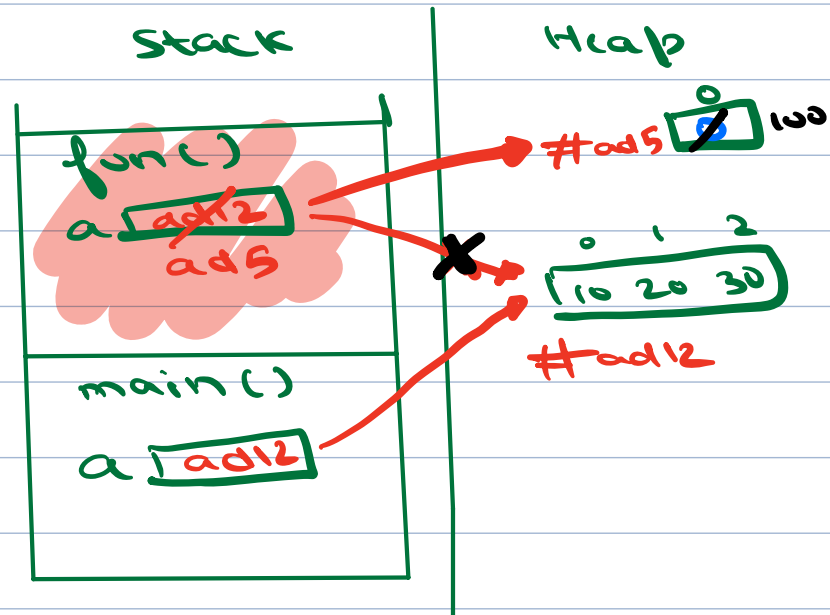
Heap changes are permanent

②

Passing reference

Quiz 3

```
static void fun(int[] a) {  
    a = new int[1];  
    a[0] = 100;  
}  
  
public static void main() {  
    int[] a = {10, 20, 30};  
    fun(a);  
    System.out.println(a[0]); 10  
}
```

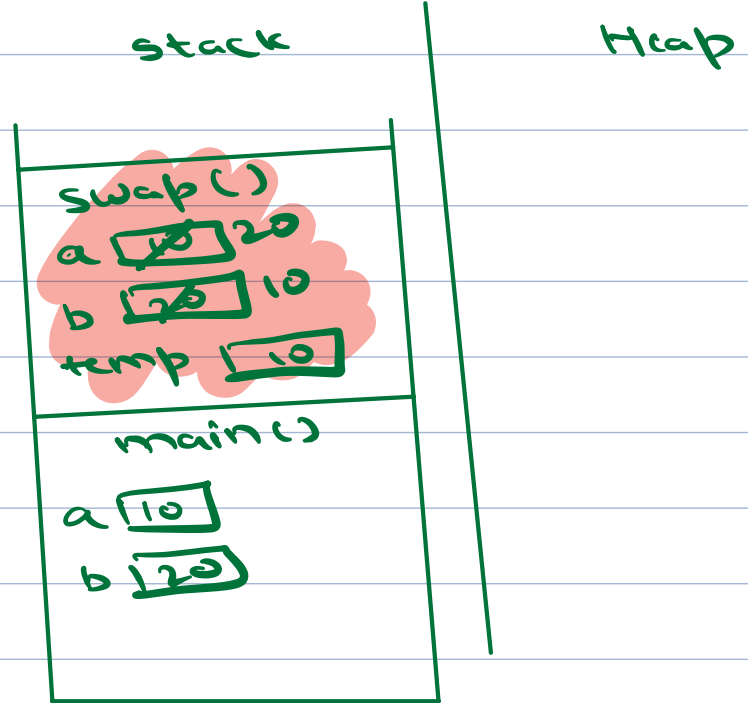


Quiz 4

```
static void swap(int a,int b) {
    int temp = a;
    a = b;
    b = temp;
}

public static void main(String args[]) {
    int a = 10;
    int b = 20;
    swap(a,b);
    System.out.println(a + " " + b);
}
```

10 20

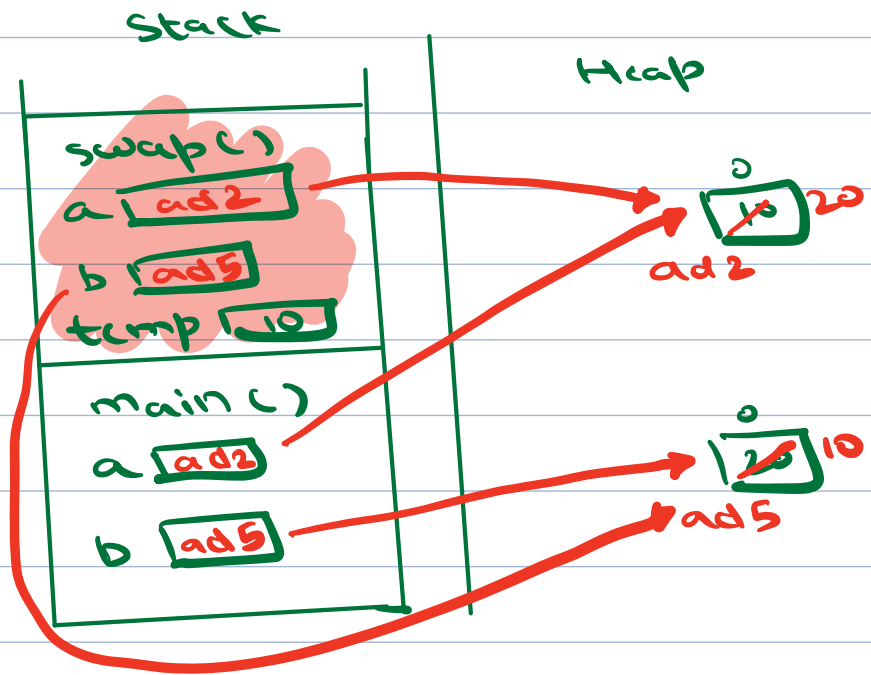


Quiz 5

```
static void swap(int[]a,int[]b) {
    int temp = a[0];
    a[0] = b[0];
    b[0] = temp;
}

public static void main(String args[]) {
    int[]a = {10};
    int[]b = {20};
    swap(a,b);
    System.out.println(a[0] + " " + b[0]);
}
```

20 10

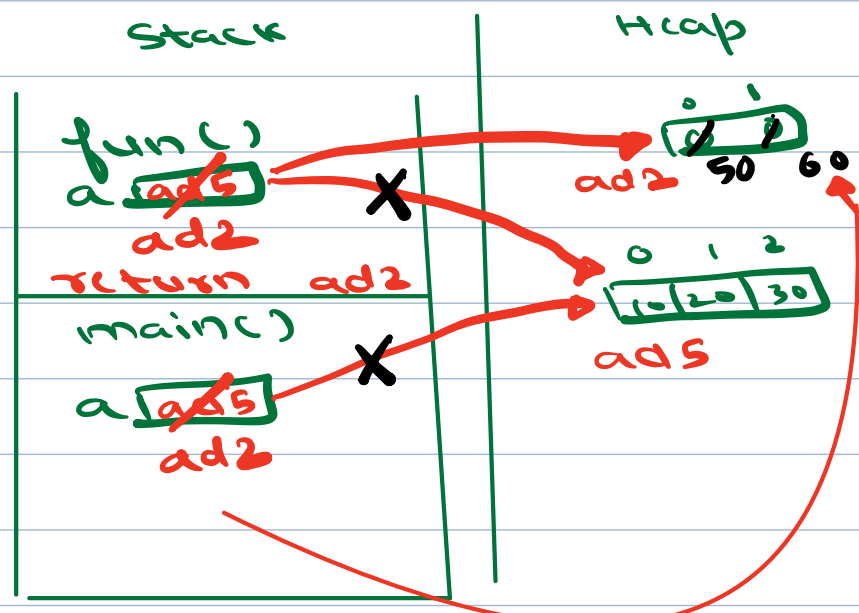


Quiz 6

```
static int[] fun(int[]a) {
    a = new int[2];
    a[0] = 50; a[1] = 60;
    return a;
}

public static void main(String args[]) {
    int[]a = {10,20,30};
    a = fun(a);
    System.out.println(a[0]);
}
```

50



Sorting : Arrangement of data in a particular order based on some parameters

2	3	9	12	17	19	Sorted in ascending order (based on magnitude)
19	6	5	2	-1	-19	Sorted in descending order (based on magnitude)
cnt of factors	1	13	9	6	12	Sorted in asc order (based on factors cnt)
	↓	↓	↓	↓	↓	
	1	2	3	4	6	

- ① Default → asc order based on value
- ② Custom Sort

Why Sorting ? Searching becomes easier
Analysis
Readability / Presenting

Java	Arrays.sort(arr) → Array	TC: $O(N \log_2 N)$
	Collections.sort(arr) → ArrayList	SC: $O(N)$
C++	sort(arr, arr+n)	→ Array
	sort(arr.begin(), arr.end())	→ Vector
python	arr.sort()	

2. In an array of N element, find count of noble integers.

Distinct

$A[i]$ is noble if

count of elements $< A[i] = A[i]$

Noble element = total count of smaller elements of current element should equal of current element for ex- $ex[0] = 1$ and if you count there is two smaller elements than i which are $(-5$ and $-10)$ and now is count $= 2$ is equal to 1 not so it is not Noble element

	0	1	2	3	4	5	
Ex	1	-5	3	5	-10	4	ans = 3
		↓	↓	↓	↓	↓	
cnt	2	1	3	5	0	4	

	0	1	2	3	
Ex	-3	0	2	5	ans = 1
	↓	↓	↓	↓	
cnt	0	1	2	3	

Brute Force: For every ele, check if its noble
↓
count smaller elements

```
int ans = 0
```

```
for (i = 0; i < n; i++) {  
    // hey, is  $A[i]$  noble?
```

```
    int cnt = 0
```

```
    for (j = 0; j < n; j++) {  
        if ( $A[j] < A[i]$ )  
            cnt++
```

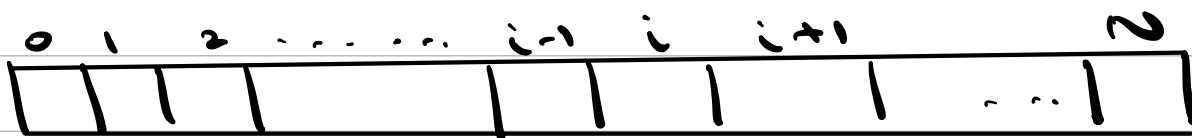
```
    }  
    if ( $A[i] == cnt$ ) ans++
```

```
return ans
```

TC: $O(N^2)$

SC: $O(1)$

2. sort data (asc)



smaller elements than $A[i]$
 $[0 \rightarrow i-1] = i$ elements

$\Rightarrow$ cnt of smaller elements $= i$

so the total count of smaller elements of i is equal to $a[i]$ element for example $a[1, -5, 3, 5, -10, 4]$ count the total smaller elements of each with and without sorting when you count after sorting then you will observe that $i = a[i]$ will come

① sort data (asc) $\rightarrow N \log N$

② $\text{int ans} = 0$
 for $i=0; i < n; i++$ $\leftarrow N$
 // hey, is $A[i]$ noble?

if $(A[i] == i) <$
 $\text{ans}++$

return ans

$N \log N + N$

TC: $O(N \log_2 N)$

SC: $O(\text{sorting algo})$

What if there are duplicates?

Ex

	0	1	2	3	4	
	-10	1	1	3	100	ans = 3
cnt	0	1	1	3	4	

For some elements, idx is not reflecting

actual count of smaller elements.

Ex	0	1	2	3	4	5	6	7	8
	-10	1	1	2	4	4	4	8	10
cnt	0	1	1	3	4	4	4	7	8

ans = 5

Index fails from 2nd occurrence of duplicates



① sort data (asc)

② int ans = 0
int cnt = 0

// cnt of noble ele
// cnt of smaller ele

if (A[0] == 0) ans++

for (int i = 1; i < n; i++) {

if (A[i] != A[i-1]) {
cnt = i

if (A[i] == cnt) ans++

return ans

TC: $O(N \log_2 N)$

SC: $O(\text{sorting algo})$

	0	1	2	3	4	5	6	7	8
Ex	-10	1	1	2	4	4	4	8	10
cnt →	0	→ 1	→ 1	→ 3	→ 4	→ 4	→ 4		

ans = ~~0~~ ~~1~~ ~~4~~ ~~7~~ ~~4~~ 5

1. Given an array of N integers, we've to delete all elements of the array. Before deleting an element, pay cost = sum of elements in the array (at that point). Find min cost.

Ex $[2, 1, 4]$

ans = 11

$2 \rightarrow 1 \rightarrow 4$

delete 2
delete 1
delete 4

$$2 + 1 + 4 = 7 +$$

$$1 + 4 = 5 +$$

$$4 = 4$$

$$\underline{\underline{16}}$$

$[2, 1, 4]$

$4 \rightarrow 2 \rightarrow 1$

delete 4
delete 2
delete 1

$$2 + 1 + 4 = 7$$

$$2 + 1 = 3 +$$

$$1 = 1$$

$$\underline{\underline{11}}$$

$[4, 6, 1]$

ans = 17

$6 \rightarrow 4 \rightarrow 1$

delete 6
delete 4
delete 1

$$4 + 6 + 1 = 11$$

$$4 + 1 = 5 +$$

$$1 = 1 +$$

$$\underline{\underline{17}}$$

Order of deletion for min cost

↓
biggest → smallest

[a] [b] [c] [d]

a d
10 2
or
2 10

Delete a $a + b + c + d$

Delete b $b + c + d$

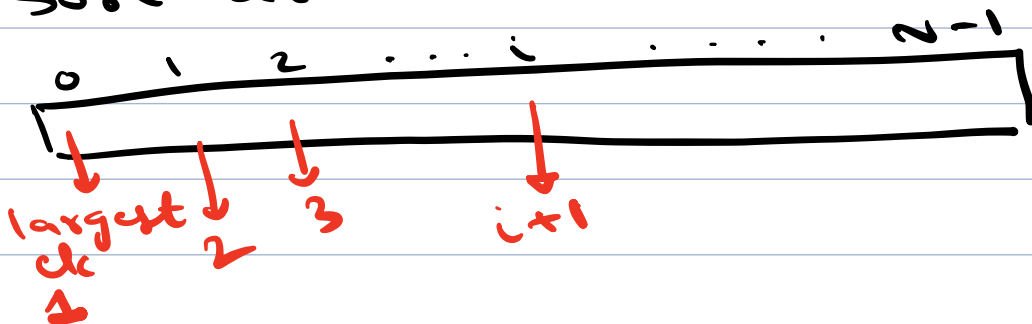
Delete c $c + d$

Delete d d

$$\underline{a + 2b + 3c + 4d}$$

$$a > b > c > d$$

Sort arr → desc order



① reverse-sort (arr)

② int cost = 0

for (i=0 ; i < n ; i++) {

cost = cost + (A[i] * (i+1))

return cost

TC: $O(N \log_2 N)$
SC: $O(1)$ (sorting algo)

[4, 9, 100, -1]

↓

[1, 2, 3]

[-1, 100, 200]

Lectcode

Striver

SDE
Sheet

A-2
Sheet